

Computing the Gradient for Logistic Regression

Math 3180

Binary Logistic Regression: Setup

Log-Likelihood

$$L = Y^T \log(\sigma(XW)) + (1 - Y)^T \log(1 - \sigma(XW))$$

Dimensions:

- ▶ X is $N \times K$ (data matrix)
- ▶ W is $K \times 1$ (weight vector)
- ▶ Y is $N \times 1$ (labels: 0s and 1s)

The Sigmoid Function

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Its derivative:

$$\begin{aligned}\frac{d}{dx}\sigma(x) &= -\frac{1}{(1 + e^{-x})^2} \cdot (-e^{-x}) \\ &= \left(\frac{e^{-x}}{1 + e^{-x}}\right) \left(\frac{1}{1 + e^{-x}}\right) \\ &= \sigma(x)(1 - \sigma(x))\end{aligned}$$

Component Form of $\sigma(XW)$

$\sigma(XW)$ is an N -vector whose i -th entry is

$$\sigma(XW)_i = \sigma \left(\sum_{j=1}^K x_{ij} w_j \right).$$

Therefore the first term in L expands as

$$Y^T \log \sigma(XW) = \sum_{i=1}^N y_i \log \sigma \left(\sum_{j=1}^K x_{ij} w_j \right).$$

Gradient: Differentiating the First Term

Applying the chain rule to $\frac{\partial}{\partial w_s}$:

$$\begin{aligned} & \frac{\partial}{\partial w_s} \sum_{i=1}^N y_i \log \sigma \left(\sum_{j=1}^K x_{ij} w_j \right) \\ &= \sum_{i=1}^N y_i \cdot \frac{1}{\sigma \left(\sum_{j=1}^K x_{ij} w_j \right)} \cdot \sigma \left(\sum_{j=1}^K x_{ij} w_j \right) (1 - \sigma \left(\sum_{j=1}^K x_{ij} w_j \right)) \cdot x_{is} \\ &= \sum_{i=1}^N y_i \left(1 - \sigma \left(\sum_{j=1}^K x_{ij} w_j \right) \right) x_{is} \end{aligned}$$

Gradient: Differentiating the Second Term

$$\begin{aligned}& \frac{\partial}{\partial w_s} (1 - Y)^T \log(1 - \sigma(XW)) \\&= \frac{\partial}{\partial w_s} \sum_{i=1}^N (1 - y_i) \log\left(1 - \sigma\left(\sum_{j=1}^K x_{ij} w_j\right)\right) \\&= \sum_{i=1}^N (1 - y_i) \cdot \frac{-\sigma\left(\sum_{j=1}^K x_{ij} w_j\right) (1 - \sigma\left(\sum_{j=1}^K x_{ij} w_j\right))}{1 - \sigma\left(\sum_{j=1}^K x_{ij} w_j\right)} \cdot x_{is} \\&= - \sum_{i=1}^N (1 - y_i) \sigma\left(\sum_{j=1}^K x_{ij} w_j\right) x_{is}\end{aligned}$$

Combining Both Terms

$$\begin{aligned}\frac{\partial L}{\partial w_s} &= \sum_{i=1}^N y_i (1 - \sigma(XW)_i) x_{is} - \sum_{i=1}^N (1 - y_i) \sigma(XW)_i x_{is} \\&= \sum_{i=1}^N y_i x_{is} - \sum_{i=1}^N y_i \sigma(XW)_i x_{is} \\&\quad - \sum_{i=1}^N \sigma(XW)_i x_{is} + \sum_{i=1}^N y_i \sigma(XW)_i x_{is} \\&= \sum_{i=1}^N y_i x_{is} - \sum_{i=1}^N \sigma(XW)_i x_{is}\end{aligned}$$

Matrix Form: Binary Case

Collecting all components $s = 1, \dots, K$ gives

$$\nabla_W L = X^T Y - X^T P = X^T (Y - P)$$

where $P = \sigma(XW)$ is the $N \times 1$ vector of predicted probabilities.

Dimension check: X^T is $K \times N$, $Y - P$ is $N \times 1$, so $\nabla_W L$ is $K \times 1$. ✓

Multiclass Classification: Setup

Log-Likelihood

$$L = \text{tr}(Y^T \log \sigma(XW))$$

where σ is now the **softmax** function (applied row-by-row), and $\text{tr}(\cdot)$ denotes the sum of diagonal entries.

Dimensions:

- ▶ X is $N \times K$ (data matrix)
- ▶ W is $K \times r$ (weight matrix)
- ▶ Y is $N \times r$ (one-hot encoding)
- ▶ XW is $N \times r$, $\sigma(XW)$ is $N \times r$

Softmax Function

For a vector $z = (z_1, \dots, z_r)$, set $F = \sum_{t=1}^r e^{z_t}$. Then

$$\sigma(z)_k = \frac{e^{z_k}}{F}.$$

Applied row-by-row to XW , the (i, t) entry of $\sigma(XW)$ is

$$\sigma(XW)_{it} = \frac{e^{z_{it}}}{F_i}, \quad z_{it} = \sum_{s=1}^K x_{is} w_{st}, \quad F_i = \sum_{t'=1}^r e^{z_{it'}}.$$

Expanding the Log-Likelihood

Using $\text{tr}(Y^T M) = \sum_{i=1}^N \sum_{t=1}^r y_{it} m_{it}$:

$$L = \sum_{i=1}^N \sum_{t=1}^r y_{it} \log \sigma(XW)_{it}.$$

Key simplification. Since $\sigma(XW)_{it} = e^{z_{it}} / F_i$:

$$\log \sigma(XW)_{it} = \log e^{z_{it}} - \log F_i = z_{it} - \log F_i.$$

Therefore

$$L = \sum_{i=1}^N \sum_{t=1}^r y_{it} z_{it} - \sum_{i=1}^N \left(\sum_{t=1}^r y_{it} \right) \log F_i = \sum_{i=1}^N \sum_{t=1}^r y_{it} z_{it} - \sum_{i=1}^N \log F_i,$$

where the last step uses $\sum_{t=1}^r y_{it} = 1$ (one-hot).

Computing $\partial L / \partial w_{pq}$

We want the (p, q) entry of the $K \times r$ gradient matrix.

$$\frac{\partial L}{\partial w_{pq}} = \sum_{i=1}^N \sum_{t=1}^r y_{it} \frac{\partial}{\partial w_{pq}} (z_{it} - \log F_i).$$

The two derivatives:

► Since $z_{it} = \sum_{s=1}^K x_{is} w_{st}$:
$$\frac{\partial z_{it}}{\partial w_{pq}} = \begin{cases} x_{ip} & t = q \\ 0 & t \neq q \end{cases}$$

►
$$\frac{\partial \log F_i}{\partial w_{pq}} = \frac{e^{z_{iq}} x_{ip}}{F_i} = \sigma(XW)_{iq} \cdot x_{ip}$$

Define the Kronecker delta $\delta_{tq} = \begin{cases} 1 & t = q \\ 0 & t \neq q \end{cases}$. Then:

$$\frac{\partial \log \sigma(XW)_{it}}{\partial w_{pq}} = x_{ip} (\delta_{tq} - \sigma(XW)_{iq}).$$

Summing Over Classes: The $t = q$ and $t \neq q$ Cases

$$\begin{aligned}\frac{\partial L}{\partial w_{pq}} &= \sum_{i=1}^N \sum_{t=1}^r y_{it} x_{ip} (\delta_{tq} - \sigma(XW)_{iq}) \\&= \sum_{i=1}^N x_{ip} \underbrace{\left(\sum_{t=1}^r y_{it} \delta_{tq} \right)}_{= y_{iq}} - \sum_{i=1}^N x_{ip} \sigma(XW)_{iq} \underbrace{\left(\sum_{t=1}^r y_{it} \right)}_{= 1} \\&= \sum_{i=1}^N x_{ip} (y_{iq} - \sigma(XW)_{iq}) = (X^T(Y - P))_{pq}\end{aligned}$$

where we used the one-hot property: $\sum_{t=1}^r y_{it} \delta_{tq} = y_{iq}$ and $\sum_{t=1}^r y_{it} = 1$.

Matrix Form: Multiclass Case

Collecting all entries (p, q) :

$$\nabla_W L = X^T (Y - P)$$

where $P = \sigma(XW)$ is the $N \times r$ matrix of softmax probabilities.

The gradient has exactly the same form as in the binary case:

	σ	W
Binary	sigmoid	$K \times 1$
Multiclass	softmax	$K \times r$

In both cases: $\nabla_W L = X^T (Y - P)$.